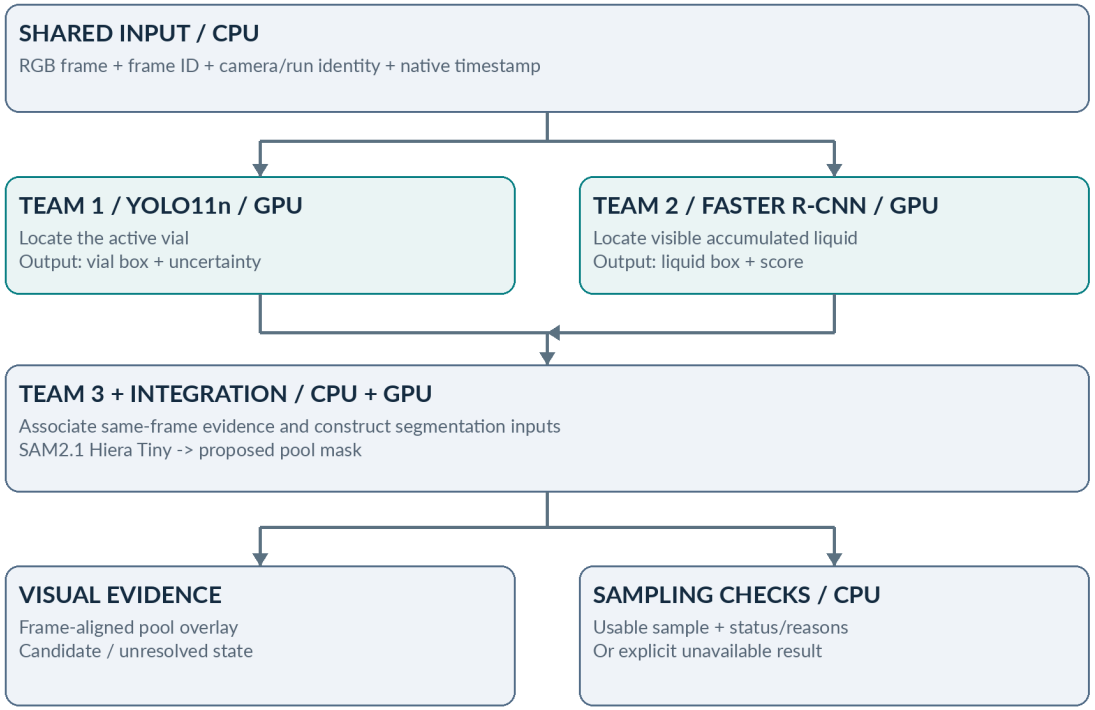
Four-frame SAM2.1 pilot

Progress review organized around the team implementation brief and shared interface contract.

SHARED ARCHITECTURE FROM THE PLANNING BRIEF



Same frame in both detector branches. Review labels are evaluation evidence, not inference inputs.

Team 3 focus: connect same-frame detections to SAM and return source-aligned full-frame masks.

READY FOR TEAM REVIEW

MODEL RUN

SAM2.1 Hiera Tiny ran on the NVIDIA GB10 using CUDA. The pilot used the Run 4 frame package and its machine-generated liquid boxes.

FRAME CONTRACT

Four source IDs and native PTS were retained at 640 x 480. Masks use original-frame coordinates and binary 0/255 PNGs.

PROMPTS

Five class-1 liquid boxes were prompted separately. The vial class was not used as a pool prompt; frame 900 records no_liquid_proposal.

HUMAN REFERENCES

Four hand-drawn masks were normalized and gap-filled for exploratory comparison. They remain draft Run 4 development labels.

CURRENT BOUNDARY

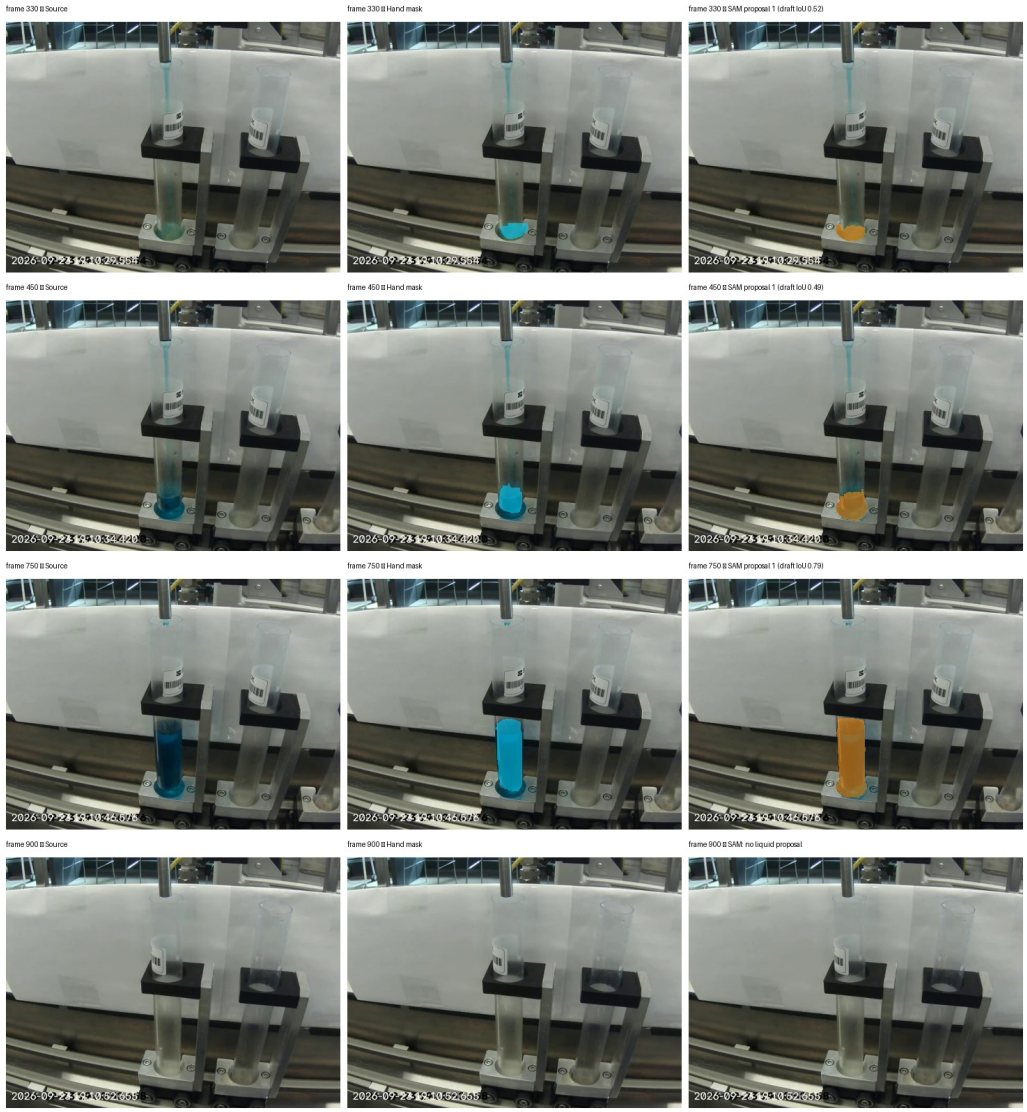
Frame-level evidence is ready. Team 1/2 detector adapter, sequence logic, and sample-usability checks are still to integrate.

DATA BOUNDARY

Runs 1-3 are the training split; Run 4 is development; Run 5 is diagnostics only. The package supplies provisional boxes, not approved pool masks or a blind final test.

Prompt comparison against draft masks

Localized boxes outperform broad boxes in this small exploratory comparison; the hand masks are not approved ground truth.



LOCALIZED BOX | PROPOSAL 1

0.599

mean IoU, n=3 present frames

Dice 0.741

BROAD BOX | PROPOSAL 2

0.085

mean IoU, n=2 prompted frames

Dice 0.156

FRAME-LEVEL OVERLAP

Frame / PTS	Proposal 1 IoU	Proposal 2 IoU	State
330 / 11.0 s	0.518	0.074	**present**
450 / 15.0 s	0.488	0.096	**present**
750 / 25.0 s	0.792	N/A	**present**
900 / 30.0 s	N/A	N/A	**absent**

WHAT THE PILOT INDICATES

Frames 330 and 450: localized lower boxes align better with the accumulated pool than the broad proposals, which over-segment.

Frame 750: proposal 1 reached IoU 0.792 / Dice 0.884 against the filled reference.
Frame 900: human absent label agrees with no proposal; no mask score applies.

Interpretation: Masks were hand-drawn, resized, stream marks excluded for the pool-only working set, and enclosed gaps filled. Scores compare prompts on Run 4; they are not SAM accuracy claims or a basis for weight tuning.

READINESS AGAINST THE IMPLEMENTATION BRIEF

What is delivered and what remains

The pilot covers mask prompting evidence. Full pipeline integration and system-level evaluation remain open.

BRIEF DELIVERABLE	CURRENT EVIDENCE	NEXT REQUIRED
READY **Data understanding**	Run 4 source IDs, PTS, 640 x 480 RGB and pixel xyxy-exclusive boxes are preserved. Machine boxes are provisional.	Keep reviewed corrections separate from inference inputs. Preserve source identity and original-size mask coordinates.
PARTIAL **Module evidence**	SAM2.1 produced five candidate masks over frames 330, 450 and 750; frame 900 has an explicit no-proposal result. Draft masks support exploratory overlap scores.	Review more varied masks. Keep accumulated pool, falling stream and glass as separate concepts.
PENDING **Integrated recording**	Not yet end-to-end: Team 1/2 model outputs are not connected to the adapter; no sequence-state or sample-usability checks are shown.	Connect same-frame detector outputs. Show early fill, established fill and transition overlays, with explicit unavailable reasons.
PARTIAL **Method explanation**	Each class-1 liquid proposal is prompted independently; vial boxes are excluded. Original-frame masks and prompt provenance are retained.	Specify behavior for oversized, inaccurate or absent prompts. Keep pool presence separate from sample reliability.
NEXT **Evaluation plan**	Only four Run 4 development examples are available. Onset delay, false alarms, temporal consistency and end-to-end runtime are not established.	Fit only with approved Run 1-3 labels. Keep Run 5 diagnostic-only; obtain a new source-disjoint recording for blind final evaluation.

NEXT MILESTONE FOR TEAM 3

1. Approve the pool/stream mask definition and add reviewed training examples from Runs 1-3. 2. Wire Team 1/2 outputs through the same-frame proposal adapter. 3. Evaluate mask quality, onset delay, false alarms, temporal consistency and runtime on new source-disjoint footage.

Artifacts: repo `team3/run_sam_pilot.py` and `team3/evaluate_masks.py`. Spark results are outside Git under `~/ai-club-tools/results/hackathon8/`.

Planning basis: Team Architecture Implementation Brief, pp. 1-3; Team Data Guide, pp. 1-3 (repository team_handoff).

Evidence: Spark pilot result and Team 3 draft-mask evaluation, Oct 22, 2026.